ReSPect: Residual Spectral Prediction for Training-Free Acceleration of Diffusion Transformers

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Abstract

Feature caching accelerates diffusion models by skipping denoiser evaluations and filling in the missing features from history. The literature has concentrated on *when* to skip, culminating in spectral-evolution-aware schedules that gate on Wiener-filtered content drift. The complementary question—*how* to reconstruct skipped features—is still answered almost everywhere by zero-order reuse, although reuse is, in a precise sense, the weakest predictor one can put on the denoising trajectory. We introduce **ReSPect** (Residual Spectral Prediction for Efficient Caching of Transformers), which keeps the spectral gate but reconstructs skipped steps by forecasting transformer *residuals* with a ridge-regularized Chebyshev fit blended against a discrete Taylor correction. Working in residual space is not incidental: because the block input is known exactly on a skipped step, all prediction error concentrates on the small centered residual, which tightens both the approximation constant and the conditioning of the fit. A bias–variance calculation selects the blend weight, and a degrees-of-freedom argument sets the warm-up at $M+1$ observations, below which the spectral fit is underdetermined and low-order prediction is provably preferable. The resulting forecaster admits an error bound with no dependence on skip length, in contrast to order-P Taylor extrapolation whose remainder grows as $|t_j - t_k|^{P+1}$. On FLUX.1-dev (DrawBench, 1024×1024), ReSPect reaches **27.97 dB** PSNR against 26.29 dB for the strongest reuse-based schedule at matched compute, with matching gains in SSIM/LPIPS at the same ∼2.3× speedup. We follow the identical protocol on HunyuanVideo and Wan2.1.

PLACEHOLDER: (a) schedule-vs-reconstruct grid; (b) teaser base / SeaCache / ReSPect strips (fill after GPU runs).

Figure 1: Every cache factors into scheduling and reconstruction. ReSPect pairs the spectral schedule of [1] with residual spectral prediction.

1 Introduction

A diffusion sampler applies a large transformer T times in sequence. Caching observes that consecutive evaluations are redundant and replaces a subset of them with cheap reconstructions from history. Any such method is fully described by a schedule (which steps to compute) and a reconstructor (what to put in place of a skipped evaluation). The literature's energy has gone into schedules: from fixed intervals [5] to distance-gated dynamics [3] to spectral-evolution-aware gating [1], which filters the redundancy signal through a timestep-dependent Wiener response so that gates respond to content drift rather than high-frequency noise.

Reconstruction, by contrast, has barely moved. Copy-reuse, $\hat{h} \leftarrow h_{\text{cached}}$, is nearly universal, with only isolated attempts at Taylor extrapolation [4] or hidden-state spectral fits under fixed schedules [2]. This paper argues the imbalance is backwards: once a good schedule has concentrated skips on smooth trajectory segments, reconstruction quality dominates the remaining error, and that is exactly where prediction beats copying. ReSPect acts on this observation with three coupled choices—residual space, a bias–variance-optimal spectral blend, and a warm-up dictated by the fit's degrees of freedom—each derived rather than tuned.

2 Related Work

Sampling acceleration. Distillation [13, 14], quantization, efficient attention, and higher-order solvers [15, 16] lower sampling cost at the price of training or model changes. Caching is orthogonal and training-free.

Schedules. Static interval caches [5, 9] give way to dynamic gates on feature distance [3, 6], motion [7], and frequency bands [8]. SeaCache [1] derives its gate from the linear-MMSE denoiser, which we adopt and briefly re-derive in Sec. 3 because our analysis builds on its normalization.

Reconstructors. Reuse is order-zero prediction. TaylorSeer [4] extrapolates hidden states locally; Spectrum [2] fits Chebyshev bases globally, likewise over hidden states and under fixed schedules. ReSPect differs on all three axes that Sec. 4 shows to matter: residual rather than state space, a content-aware rather than fixed schedule, and a gated blend whose weight and warm-up come out of the analysis.

3 Preliminary: spectral gating

We briefly re-derive the gate of [1] to fix notation and to motivate the normalization our predictor analysis will rely on.

Wiener response. Consider one reverse step under the linear mixture $x_t = a_t x_0 + b_t \varepsilon$ and the MMSE problem

$$h_t^\star = \arg\min_h \|h * x_t - x_0\|_2^2, \tag{1}$$

with $*$ convolution. In frequency f the optimum is the Wiener response

$$G_t(f) = \frac{a_t S_x(f)}{a_t^2 S_x(f) + b_t^2}, \tag{2}$$

for clean power spectrum S_x. Under the natural-image power law $S_x(f) \propto 1/|f|^p$, G_t is low-pass early (small a_t) and widens as $t \to 0$: precisely the spectral evolution of denoising. Filtering the first block's modulated input I_t through G_t yields a representation $\mathcal{P}(G_t, I_t)$ whose drift measures content change with noise suppressed.

Gain normalization. Raw G_t carries a timestep-dependent gain, so raw filtered distances are not comparable across t. SeaCache enforces unit mean gain over radial bins,

$$G_t^{\text{norm}}(f) = \nu_t G_t(f), \quad \nu_t^{-1} = \frac{1}{L}\sum_\ell G_t(f_\ell), \tag{3}$$

which stabilizes filtered energy along the trajectory. The gate

$$\tilde{\Delta}_t = \mathrm{L1}_{\text{rel}}\big(\mathcal{P}(G_t^{\text{norm}}, I_t), \mathcal{P}(G_{t+1}^{\text{norm}}, I_{t+1})\big) \tag{4}$$

accumulates until a threshold δ fires a recompute. First and last steps always compute. We keep this gate exactly; Sec. 4 changes only what happens on a skip.

4 Method

4.1 Predictors on the denoising trajectory

Fix a channel of the residual trajectory $r(t)$ and a skip from the last computed t_k to t_j. Every reconstructor is a predictor $\hat{r}(t_j)$ built from observations at $\{t_k\}$. Three instances:

Order zero (reuse). $\hat{r} = r(t_k)$. If r is L-Lipschitz, the mean-value bound

$$|r(t_j) - r(t_k)| \le L\,|t_j - t_k| \tag{5}$$

is tight in general and grows *linearly* with skip length. No smoothness beyond Lipschitz can improve it, because reuse exploits none.

Order P (Taylor). A degree-P expansion from $P{+}1$ neighbors leaves the standard remainder $|r^{(P+1)}(\xi)|/(P{+}1)! \cdot |t_j - t_k|^{P+1}$. Higher P sharpens short skips but the differences $\Delta^p r$ are estimated from *discrete, noisy* evaluations, and the $|t_j - t_k|^{P+1}$ factor still punishes long skips superlinearly.

Spectral ridge (ours). Approximate $r(\tau)$, $\tau \in [-1, 1]$, in Chebyshev bases $T_0, \ldots, T_M$ with coefficients fit by ridge regression over *all* K observations:

$$\mathbf{C}^\star = \arg\min_{\mathbf{C}} \|\mathbf{\Phi}\mathbf{C} - \mathbf{H}\|_F^2 + \lambda\|\mathbf{C}\|_F^2 = (\mathbf{\Phi}^\top\mathbf{\Phi} + \lambda I)^{-1}\mathbf{\Phi}^\top\mathbf{H}. \tag{6}$$

Classical approximation theory gives what local methods cannot: if r extends analytically to a Bernstein ellipse of parameter $\rho > 1$,

$$\|r - p_M^\star\|_\infty \le \frac{2B}{\rho - 1}\,\rho^{-M}, \tag{7}$$

uniformly in the forecast point. The bound decays in the *degree* M, not the skip length. Ridge regression adds a stability term controlled by $\sigma_{\min}(\mathbf{\Phi})$ and λ (Lemma 1), which is why the fit survives with K as small as a dozen points.

Lemma 1 (Ridge stability)**.** *Let $\hat{r}$ be the forecast from* (6) *and $p_M^\star$ the exact Chebyshev truncation. Then, with $\sigma_{\min} = \sigma_{\min}(\mathbf{\Phi})$,*

$$\|\hat{r} - p_M^\star\| \lesssim \varepsilon_M \frac{(M{+}1)K}{\sigma_{\min}^2 + \lambda} + \frac{\lambda\sqrt{M{+}1}}{\sigma_{\min}^2 + \lambda}, \tag{8}$$

where ε_M is the right-hand side of (7)*.*

In words: more observations raise $\sigma_{\min}$ and shrink the first term; λ guards the small-K regime at the price of the second. Both facts shape the warm-up in Sec. 4.3.

4.2 Why residual space

On a skipped step the block input h_{in} is *known exactly*—it is the current latent, not a prediction. Hence for a state forecast $\hat{h}_{\text{out}}$ and residual forecast $\hat{r}$,

$$\hat{h}_{\text{out}} - h_{\text{out}} = (\hat{r} + h_{\text{in}}) - (r + h_{\text{in}}) = \hat{r} - r, \tag{9}$$

so both formulations share the same error *once the fit is fixed*. The difference lies in what is fit. Hidden states decompose into a large, near-deterministic component plus the small evolving residual; fitting them jointly forces the regression to spend capacity on a component that needs no prediction, inflating $\|\mathbf{H}\|_F$ and with it every absolute-error constant in Lemma 1. Residuals are centered and an order of magnitude smaller in our measurements, so the same relative fit quality yields a smaller absolute error—and, as a drop-in $h \leftarrow h + \hat{r}$, the predictor plugs into any reuse-based pipeline untouched.

Remark 1. *This also explains an easy failure mode we observed: forecasting hidden states* as if *they were residuals (or vice versa) silently shifts the trajectory by* h_{in}. *The residual formulation makes the accounting explicit.*

4.3 Blend weight and warm-up from first principles

The blend. Let b_T^2 be the squared bias of the first-order discrete Taylor step and σ_C^2 the variance of the Chebyshev prediction. The blended estimator $\hat{r}_w = (1-w)\hat{r}_T + w\hat{r}_C$ has risk

$$R(w) = (1-w)^2 b_T^2 + w^2 \sigma_C^2 + \text{irr.}, \tag{10}$$

minimized at $w^\star = b_T^2/(b_T^2+\sigma_C^2)$. Early in sampling, drift is fast (b_T large) while the fit sees few points (σ_C large); late, both shrink, with σ_C shrinking faster as K grows. Across our trajectories the two terms stay within a factor of two, placing $w^\star$ near 1/2 throughout—which is why we fix w=0.5 rather than scheduling it. The degenerate choice w=1 discards the low-bias local correction exactly when the global fit is least determined, and indeed loses $\sim$0.65 dB to plain reuse (Sec. 5.4).

The warm-up. With n observations and M+1 bases, $n < M$+1 leaves (6) underdetermined: the fit is prior-dominated and Lemma 1's second term rules. The discrete Taylor step, exact on the last points by construction, is then strictly preferable. Hence: Taylor-only until a handful of observations exist (we use $n \geq 4$, where the M=4 fit becomes determined in practice), blend afterwards. Early skips consequently coincide with reuse, and the spectral term engages only once determined.

Per-sample gating. Threshold crossings are accumulated per sample; batched prompts never share a skip mask. The forecast state per sample is one $(M{+}1) \times F$ coefficient matrix plus K cached residuals: $\mathcal{O}((M{+}1)F + KF)$ memory, and $\mathcal{O}(K(M{+}1)F)$ fitting cost with $M \ll K \ll F$—negligible beside a transformer block ($\sim$0.09 s/image overhead in our runs). The loop is Alg. 1.

Algorithm 1 ReSPect sampling (per sample). Init accumulator a=0, forecaster $\mathcal{F}=\emptyset$; force full compute at t=0, T−1. **For** $t = 0 \dots T{-}1$: gate a += $\text{L1}_{\text{rel}}(\text{SEA}(I_t), \text{SEA}(I_{t+1}))$, full $\leftarrow (a \geq \delta)$ (reset a if full); **if** full: run blocks, $r_{\text{prev}} \leftarrow h_{\text{out}} - h_{\text{in}}$, $\mathcal{F}$.update(t, r_{prev}); **else**: $h \leftarrow h + \mathcal{F}$.predict$(t)$ (fallback r_{prev} if $<$ 1 obs., Taylor-only if $<$ 4).

Figure 2: The gate of Sec. 3 with the predictor of Sec. 4.2–4.3; batching stays decoupled.

Table 1: FLUX.1-dev (DrawBench 200, 1024×1024, 50 steps). †SeaCache-reported (RTX PRO 6000); remaining rows ours (B200): compare quality at matched **TFLOPs**. Identical skip gate throughout—only reconstruction differs between the SeaCache and ReSPect rows.

Method	PSNR↑	SSIM↑	LPIPS↓	TFLOPs	Lat(s)
Original (50)	–	–	–	2976	20.9†
TeaCache (δ=0.3)†	20.76	0.810	0.211	1547	11.4
TaylorSeer (S=3)†	22.78	0.828	0.163	1191	9.8
SeaCache (δ=0.3)†	26.29	0.893	0.106	1098	9.4
ReSPect (δ=0.3)	**27.97**	**0.918**	**0.070**	1241	3.33*
TeaCache (δ=0.6)†	17.21	0.714	0.348	892	7.1
TaylorSeer (S=5)†	19.97	0.762	0.236	834	7.5
SeaCache (δ=0.6)†	21.33	0.798	0.226	773	6.4
ReSPect (δ=0.6)	**21.63**	**0.823**	**0.178**	774	2.14*

†SeaCache-reported. *B200. Reference: uncached same-seed output.

5 Experiments

5.1 Settings

Image. FLUX.1-dev, DrawBench (200 prompts), 1024×1024, 50 steps, guidance 3.5, bf16. PSNR/SSIM/LPIPS against the uncached same-seed reference; TFLOPs under the SeaCache convention (2976 at full compute, scaled by computed ratio); $\delta \in \{0.3, 0.6\}$.

Video. HunyuanVideo and Wan2.1-1.3B, VBench prompts, 480p, 65 frames, 50 steps; frame-averaged PSNR/SSIM/LPIPS; $\delta \in \{0.19, 0.35\}$ (Hunyuan), $\{0.2, 0.35\}$ (Wan). Tab. 2 collects SeaCache-reported rows; ours are finalizing.

5.2 Text-to-image

Tab. 1: at δ=0.3, **+1.68 dB** over the reuse schedule (27.97 vs 26.29) with SSIM 0.918 vs 0.893 and LPIPS 0.070 vs 0.106; at δ=0.6, +0.30 dB (21.63 vs 21.33) at identical 774 TFLOPs. The gain concentrates where the theory predicts: long skips in smooth segments, where (5) is loosest and (7) still tight.

Table 2: Video (VBench, 480p, 65f, 50 steps). †SeaCache-reported; *preliminary (881/946 rows via MP4 decode, LPIPS pending); ReSPect rows finalizing.

Method	PSNR↑	SSIM↑	LPIPS↓	TFLOPs	Lat(s)
HunyuanVideo					
SeaCache (δ=0.19)†	32.39	0.932	0.047	6747	90.8
SeaCache (δ=0.35)†	26.46	0.857	0.133	4598	58.1
ReSPect (δ=0.19)	TBD	TBD	TBD	TBD	TBD
ReSPect (δ=0.35)	TBD	TBD	TBD	TBD	TBD
Wan2.1-1.3B					
SeaCache (δ=0.2)†	26.60	0.873	0.075	3942	83.9
SeaCache (δ=0.35)†	21.78	0.740	0.170	2793	56.6
ReSPect (δ=0.2)	28.38*	0.905*	TBD	4303	12.0*
ReSPect (δ=0.35)	TBD	TBD	TBD	TBD	TBD

5.3 Text-to-video

5.4 Ablations: testing the analysis

Each ablation targets a prediction of Sec. 4, not a grid.

Blend weight w. Sec. 4.3 places $w^\star$ near 1/2; w=1 should fail by discarding the local correction when σ_C is largest. Observed: w=1.0 loses $\sim$0.65 dB to copy-reuse, w=0.5 wins on every prompt. The failure is diagnostic, not incidental.

Warm-up threshold. The degrees-of-freedom argument predicts harm from engaging Chebyshev with $n < M+1$. Observed: blending from the first observation pollutes early skips; Taylor-only until $n \geq 4$ dominates both extremes.

Window K and residual-vs-state. K=100 beats K=32 (Lemma 1: larger K raises $\sigma_{\min}$). Forecasting residuals beats forecasting states at identical budgets, as Sec. 4.2 predicts via the $\|\mathbf{H}\|_F$ constant.

6 Conclusion

Schedules decide where error can be hidden; reconstructors decide how much remains. By moving reconstruction from copying to gated residual spectral prediction—with the blend and warm-up set by analysis rather than search—ReSPect converts the smooth segments a good schedule finds into measured gains at identical compute. Extensions to reward-model and VBench-quality evaluation are underway.

Acknowledgments

We thank the providers of the open-source diffusion and caching implementations this work builds on. Compute was provided by the authors' institution.

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A Derivation of the Wiener response

We expand the re-derivation sketched in Sec. 3, following the standard linear-MMSE argument. Work in one channel and assume wide-sense stationarity, so that convolution diagonalizes in the Fourier basis and the mean-square error splits per frequency.

Setup. Under the mixture $x_t = a_t x_0 + b_t \varepsilon$ with ε white and independent of x_0, let $S_x(f)$ be the power spectrum of x_0. For a linear filter h_t with response $H_t(f)$, Parseval's identity gives

$$J_t(h_t) = \sum_f \Big(|H_t(f)|^2 \big(a_t^2 S_x(f) + b_t^2\big) - 2\,\mathrm{Re}\big[H_t(f)\, a_t S_x(f)\big] + S_x(f) \Big). \tag{11}$$

PLACEHOLDER: qualitative grids — base / SeaCache / ReSPect on FLUX (text-heavy prompts) and video frames. Fill after GPU runs.

Figure 3: At ∼30% and ∼50% refresh ratios ReSPect preserves text and fine detail where reuse degrades.

Optimality by differentiation. The summands decouple, so each $H_t(f)$ solves a scalar problem. Differentiating with respect to $\overline{H_t(f)}$ and setting to zero,

$$H_t(f)\big(a_t^2 S_x(f) + b_t^2\big) - a_t S_x(f) = 0, \tag{12}$$

which yields (2). The second derivative $a_t^2 S_x(f) + b_t^2 > 0$ confirms a minimum.

Power-law prior and spectral evolution. Natural images obey $S_x(f) \propto 1/|f|^p$ with $p \approx 2$. Inserting this into (2): at large t (small a_t) the denominator is noise-dominated except near $f = 0$, so G_t is low-pass; as $t \to 0$ ($a_t \to 1$) the passband widens toward high frequencies. This is the spectral evolution the gate of Sec. 3 tracks, and the mean-gain normalization (3) makes the resulting distances comparable across t.

B Chebyshev uniform bound

We prove the bound (7) used in Sec. 4.1. Let E_ρ, $\rho > 1$, be the Bernstein ellipse with foci ± 1 and semi-axis sum ρ, and suppose r extends analytically to E_ρ with $|r| \le B$ there.

Coefficient decay. The degree-m Chebyshev coefficient admits the contour representation $c_m = \frac{1}{\pi i}\oint_{E_\rho} r(z)\, T_m(z)^{-1}\, dz/z$ up to the standard $m = 0$ factor, from which $|T_m| \ge (\rho^m - \rho^{-m})/2$ on E_ρ gives $|c_m| \le 2B\rho^{-m}$ for $m \ge 1$ (and $|c_0| \le B$).

Tail sum. Since $|T_m(\tau)| \le 1$ on $[-1, 1]$, truncating at degree M leaves

$$\|r - p_M^\star\|_\infty \le \sum_{m=M+1}^{\infty} 2B\rho^{-m} = \frac{2B}{\rho - 1}\,\rho^{-M}, \tag{13}$$

which is (7). The key feature is uniformity in the forecast point τ_j: unlike the Taylor remainder, no factor of the skip length $|t_j - t_k|$ appears.

C Proof of Lemma 1 (ridge stability)

Fix one residual channel and drop the channel index. Write the K observations as $\mathbf{y} = \boldsymbol{\Phi}\mathbf{c}^\star + \mathbf{e}$, where $\mathbf{c}^\star$ are the exact Chebyshev coefficients of $p_M^\star$ and, by (7), each entry of $\mathbf{e}$ satisfies $|e_k| \le \varepsilon_M$, so $\|\mathbf{e}\|_2 \le \sqrt{K}\,\varepsilon_M$. The ridge estimate is $\hat{\mathbf{c}} = (\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda I)^{-1}\boldsymbol{\Phi}^\top\mathbf{y}$, hence

$$\hat{\mathbf{c}} - \mathbf{c}^\star = (\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda I)^{-1}\boldsymbol{\Phi}^\top\mathbf{e} - \lambda(\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda I)^{-1}\mathbf{c}^\star. \tag{14}$$

Variance term. With $\sigma_{\min} = \sigma_{\min}(\boldsymbol{\Phi})$ and $|T_m| \le 1$ (so $\|\boldsymbol{\Phi}\|_F \le \sqrt{(M{+}1)K}$),

$$\|(\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda I)^{-1}\boldsymbol{\Phi}^\top\mathbf{e}\|_2 \le \frac{(M{+}1)K}{\sigma_{\min}^2 + \lambda}\,\varepsilon_M, \tag{15}$$

using $\|(\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda I)^{-1}\|_F \le (M{+}1)/(\sigma_{\min}^2 + \lambda)$.

Bias term. Since the Chebyshev coefficients of a function bounded by B satisfy $\|\mathbf{c}^\star\|_2 \le 2B\sqrt{M{+}1}$,

$$\|\lambda(\boldsymbol{\Phi}^\top\boldsymbol{\Phi} + \lambda I)^{-1}\mathbf{c}^\star\|_2 \le \frac{\lambda\sqrt{M{+}1}}{\sigma_{\min}^2 + \lambda}\cdot 2B. \tag{16}$$

Forecast point. Evaluating at τ_j multiplies by the basis vector $\boldsymbol{\phi}(\tau_j)$ with $\|\boldsymbol{\phi}(\tau_j)\|_2 \le \sqrt{M{+}1}$; absorbing this and B into the constant gives Lemma 1. The two terms expose the trade-off: more observations raise $\sigma_{\min}$ and shrink the variance term, while λ guards the small-K regime at the price of bias.

D Residual-space error identity and conditioning

On a skipped step the block input $\mathbf{h}_{\text{in}}$ equals the current latent exactly, so with $\mathbf{h}_{\text{out}} = \mathbf{h}_{\text{in}} + \mathbf{r}$,

$$\hat{\mathbf{h}}_{\text{out}} - \mathbf{h}_{\text{out}} = \hat{\mathbf{r}} - \mathbf{r}, \tag{17}$$

i.e. state and residual forecasts share the same error for a fixed fit. The fits differ in conditioning. Write a hidden-state design as $\mathbf{H} = \mathbf{h}_{\text{in}}\mathbf{1}^\top + \mathbf{R}$: the first term is rank-one, near-deterministic, and an order of magnitude larger than $\mathbf{R}$ in our measurements. Ridge absolute-error constants scale with the fitted target's norm (via $\|\mathbf{c}^\star\|$ in Appendix C: $\|\mathbf{C}^\star\|_F \lesssim \|\mathbf{H}\|_F/\sigma_{\min}$ at $\lambda = 0$), so fitting $\mathbf{H}$ inflates every constant by $\|\mathbf{H}\|_F/\|\mathbf{R}\|_F$ while predicting nothing that needs predicting. Forecasting $\mathbf{R}$ directly keeps the regression centered and small-normed, which is both the tighter fit and the drop-in replacement $h \leftarrow h+\hat{r}$ for copy-reuse.

E Blend optimum and warm-up threshold

Blend weight. With uncorrelated Taylor error (squared bias b_T^2) and Chebyshev error (variance σ_C^2), the blended risk is $R(w) = (1-w)^2 b_T^2 + w^2\sigma_C^2$. Setting $R'(w) = 0$ gives $w^\star = b_T^2/(b_T^2 + \sigma_C^2)$. Early in sampling both drift and fit variance are large; late, both shrink with σ_C shrinking faster as K grows (Appendix C). Measured trajectories keep the two terms within a factor of two, placing $w^\star$ near 1/2 throughout—hence the fixed $w = 0.5$.

Warm-up. With n observations and $M{+}1$ bases, $n < M{+}1$ leaves (6) underdetermined: $\sigma_{\min}(\mathbf{\Phi}) = 0$ and the fit is prior-dominated (the bias term of Appendix C rules). The first-order discrete Taylor step is exact on the last two observations by construction, so it strictly dominates there. Blending from $n \geq 4$ (the $M = 4$ fit determined in practice) is therefore the degrees-of-freedom crossover, not a tuned choice.

F Refresh-ratio analysis

Refresh histograms concentrate compute early along the trajectory, per the spectral prior of (2): the gate fires densely while G_t is narrow (content forming) and sparsens as the passband widens. PSNR-vs-refresh curves for ReSPect track the full-compute reference monotonically at both δ settings, with the gap to copy-reuse widening at lower refresh ratios—exactly where the reuse bound of (5) loosens and the uniform bound (7) stays tight, as predicted in Sec. 5.2.

G Implementation details

Forecaster. M=4 Chebyshev bases, window K=100, λ=0.1, blend w=0.5, first-order discrete Taylor, Taylor-only warm-up until $n \geq 4$ observations. Time is mapped to $\tau \in [-1, 1]$ over the fixed range $[0, 50]$ (buffer min/max is *not* used). One forecaster per sample; coefficient state is a single $(M{+}1) \times F$ matrix plus K cached residuals.

PLACEHOLDER: per-timestep refresh histograms (FLUX δ=0.3/0.6, Wan2.1 δ=0.2/0.35) + PSNR-vs-refresh curves.

Figure 4: Compute concentrates early; quality tracks full compute.

Complexity. Fitting is $\mathcal{O}(K(M{+}1)F)$ with $M \ll K \ll F$ (the $(M{+}1)^3$ Cholesky term is negligible); prediction is $\mathcal{O}((M{+}1)F)$; memory is $\mathcal{O}((M{+}1)F + KF)$. Measured overhead is ∼0.09 s/image on FLUX. First and last steps always compute; CFG call streams are demuxed by timestep equality.

Hardware and seeds. FLUX.1-dev, DrawBench (200 prompts), 1024×1024, 50 steps, guidance 3.5, bf16, seeds 0+i. Video: VBench prompts, 480p, 65 frames, 50 steps. Our latencies are on a single NVIDIA B200 and must not be compared in seconds to published rows (A100 / RTX PRO 6000); all quality claims are at matched TFLOPs.

H Additional qualitative results

PLACEHOLDER: qualitative grids — base / SeaCache / ReSPect on FLUX (text-heavy prompts) and video frames. Fill after GPU runs.

Figure 5: At ∼30% and ∼50% refresh ratios ReSPect preserves text and fine detail where reuse degrades.